Jake Dearborn

Computational Biologist
Machine Learning for Genomics and Transcriptomics

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Professional Summary

Computational biologist and machine learning researcher developing interpretable models for genomic sequence and transcriptomic data. My PhD work focuses on long-context sequence-to-function modeling, genomic foundation model adaptation, and computational dissection of regulatory programs underlying RNA processing and immune cell type-specific alternative splicing. I develop end-to-end computational workflows spanning RNA-seq label generation, deep learning model development, attribution-based interpretation, and large-scale training on GPU/HPC systems, with prior industry experience in regulated assay development and sequencing-based experimental workflows.

Technical Expertise

- **Machine Learning & Modeling:** PyTorch, PyTorch Lightning, transformers, genomic foundation models, multitask learning, LoRA/PEFT, representation learning, model interpretability, attribution methods, Optuna, Weights & Biases
- **Computational Genomics & Transcriptomics:** bulk RNA-seq, scRNA-seq, CITE-seq, alternative splicing analysis, PSI / intron retention modeling, transcript isoform analysis, regulatory genomics, multi-omics integration
- **Bioinformatics Tools & Data Processing:** STAR, Salmon, StringTie, rMATS, Seurat, Scanpy, genomic interval analysis, HDF5/NPZ workflows, custom label-generation pipelines, reproducible Python/R/Bash environments
- **Scientific Computing & Infrastructure:** Python, R, Bash, Linux, Git/GitHub, Conda, Docker, SLURM, mixed-precision GPU training, HPC workflow engineering
- **Experimental & Assay Experience:** oligonucleotide design, targeted capture, RNA library preparation, assay optimization, sequencing workflow development, qPCR/PCR, gel electrophoresis, GxP assay validation

Software & Research Tools

- **gReLU contribution:** Contributed a feature addition via pull request enabling TF-MoDISco runs from precomputed attribution scores within the gReLU regulatory genomics framework.
- **Sequence modeling tooling:** Developed reusable training, evaluation, and experiment-management utilities for biological sequence models, including data preparation, configuration, and model-assessment workflows.
- **Analysis automation:** Built Python and R tools to standardize QC, automate result reporting, and generate analysis-ready datasets across sequencing-based projects.

Research & Professional Experience

University of Vermont — Burlington, VT
PhD Researcher, Molecular Biology — Sep 2021 – Present
PhD Candidate, defending June 2026

- Architected machine learning models and computational workflows to study RNA biology, regulatory genomics, and immune cell type-specific alternative splicing
- Engineered and trained long-context sequence-to-function models, including transformer-based architectures and adapted genomic foundation models (e.g., Borzoi), to predict splicing outcomes from genomic

sequence

- Built end-to-end label-generation workflows integrating bulk RNA-seq, transcript assembly, and alternative splicing quantification to produce training datasets and regulatory annotations for deep learning
- Implemented and optimized scalable PyTorch/PyTorch Lightning pipelines for GPU and HPC environments, including mixed-precision training, experiment tracking, and hyperparameter tuning
- Developed and applied interpretation workflows using attribution and motif discovery tools (DeepSHAP, TF-MoDISco, MEME/FIMO) to identify candidate cis-regulatory features learned by neural networks
- Spearheaded single-cell and multi-omic transcriptomic analyses across Drop-seq, 10x Genomics, PIP-seq, and CITE-seq datasets, translating findings into biological insights, figures, and manuscripts
- Designed data-splitting and evaluation strategies to mitigate signal sparsity, correlated outputs, and biological data leakage in high-dimensional genomic modeling tasks
- Collaborated with experimental researchers to refine sequencing designs, guide hypothesis generation, and translate computational results into experimentally testable models of RNA regulation

Haematech Biopharma Services (now Prolytix) Essex Junction, VT
QC Specialist / R&D Associate Sep 2018 – Aug 2021

- Executed GxP-compliant assay development, validation, and quality control workflows in a regulated biopharma environment supporting cell therapy and related laboratory processes.
- Analyzed assay performance data to assess reproducibility, sensitivity, and sources of variability, supporting troubleshooting, optimization, and validation readiness.
- Contributed to method transfer, documentation, and quality systems aligned with regulated laboratory standards.
- Worked at the interface of experimental execution, data review, and process robustness in clinically oriented assay workflows.

Lab Technician Sep 2015 – Sep 2018
University of Vermont

- Supported molecular biology experiments, laboratory operations, instrumentation, and research infrastructure in an academic lab setting
- Contributed to experimental troubleshooting, study execution, and training of junior personnel

Publications

First / Co-First Author

1. **Dearborn J**, Frankiw L, Limoge DW, Burns CH, Vlach L, Turpin P, Kirch T, Miller ZD, Dowell W, Languon S, *et al.* (2026). *Programmed delayed splicing: A mechanism for timed inflammatory gene expression*. Reviewed Preprint *eLife*. doi:10.1101/443796

2. Dowell W, **Dearborn J**, Languon S, Miller Z, Kirch T, Paige S, Garvin O, Kjendal L, Harby E, Zuchowski AB, *et al.* (2024). *Distinct inflammatory programs underlie the intramuscular lipid nanoparticle response*. *ACS Nano*. doi:10.1021/acsnano.4c08490

3. Gasek N, **Dearborn J**, Enes SR, Pouliot R, Louie J, Phillips Z, Wrenn S, Uhl FE, Riveron A, Bianchi J, *et al.* (2021). *Comparative immunogenicity of decellularized wild type and* α*1,3-galactosyltransferase knockout pig lungs*. *Biomaterials*. doi:10.1016/j.biomaterials.2021.121029

Co-Author

1. Kjendal L, Whelihan C, Garvin O, Will B, Lee I, Miller ZD, Dowell W, **Dearborn J**, Languon S, Kirch T, *et al.* (2026). *Iterative immunoprecipitation and phage pre-wash dramatically improve epitope-resolved*

serology by VirScan. *Frontiers in Virology*. doi:10.3389/fviro.2026.1761741

2. Snoke DB, van der Velden JL, Bellafleur ER, **Dearborn J**, Lenahan SM, Beal AE, Aboushousha R, Heininger SCJ, Ather JL, Mank MM, *et al.* (2025). *Early adipose tissue wasting in a preclinical model of human lung cancer cachexia*. *Cell Reports*. doi:10.1016/j.celrep.2025.116278

3. Gasek N, Park HE, Uriarte JJ, Uhl FE, Pouliot RA, Riveron A, Moss T, Phillips Z, Louie J, Sharma I, **Dearborn J**, *et al.* (2021). *Development of alginate and gelatin-based pleural and tracheal sealants*. *Acta Biomaterialia*. doi:10.1016/j.actbio.2021.06.048

4. Abreu SC, Hampton TH, Hoffman E, **Dearborn J**, Ashare A, Sidhu KS, Matthews DE, McKenna DH, Amiel E, Barua J, *et al.* (2020). *Differential effects of the cystic fibrosis lung inflammatory environment on mesenchymal stromal cells*. *American Journal of Physiology - Lung Cellular and Molecular Physiology*. doi:10.1152/ajplung.00218.2020

5. Abreu SC, Rolandsson Enes S, **Dearborn J**, Goodwin M, Coffey A, Borg ZD, Dos Santos CC, Wargo MJ, Cruz FF, Loi R, *et al.* (2019). *Lung inflammatory environments differentially alter mesenchymal stromal cell behavior*. *American Journal of Physiology - Lung Cellular and Molecular Physiology*. doi:10.1152/ajplung.00263.2019

6. Wrenn SM, Griswold ED, Uhl FE, Uriarte JJ, Park HE, Coffey AL, **Dearborn J**, Ahlers BA, Deng B, Lam YW, *et al.* (2018). *Avian lungs: A novel scaffold for lung bioengineering*. *PLOS One*. doi:10.1371/journal.pone.0198956

Preprints / Submitted

1. **Dearborn J**, Stagg R, Kirch T, Miller Z, Languon S, Dowell W, Limoge D, Majumdar D. *Learning and probing sequence representations of immune cell alternative splicing.* In preparation.

2. Miller Z, **Dearborn J**, Barrantes-Reynolds R, Paculova H, Honson D, Sha J, Deng W, Kirch T, Dowell W, Languon S, Freeman K, Frietze S, Wohlschlegel J, Majumdar D. *PABPC1 modulates PA site choice through direct interaction with immunoglobulin pre-mRNA*. Under review, *Nucleic Acids Research*.

Selected Presentations

- **Programmed Delayed Splicing: A Mechanism for Timed Inflammatory Gene Expression.** Keystone Symposium: AI in Molecular Biology, Poster, 2025.
- **Uncovering Splice Regulator Landscapes with Genomics Modeling Approaches.** Research in Progress Seminar, Vermont Collaborative for Immunity and Host-Microbe Interactions, 2025.
- **Implementing and Interpreting Transformer Models in Genomics.** Guest Lecture, Deep Learning in Biology (Graduate Course), University of Vermont, 2025.
- **The Mechanics and Utility of Transformer Models in Genomics.** Guest Lecture, Deep Learning in Biomedical Engineering (Undergraduate Course), University of Vermont, 2024.
- **Understanding Features Regulating Intron Retention in Inflammatory Genes.** Research in Progress Seminar, Cell and Molecular Biomedical Sciences Program, 2025.
- **Beyond the Horizon: Exceeding Technological Boundaries in Immune Repertoire Analysis.** Three Minute Thesis Elevator Pitch Competition, University of Vermont, 2024.

Education

University of Vermont Burlington, VT
Ph.D. Candidate, Molecular Biology Expected June 2026

University of Vermont Burlington, VT
B.S., Neuroscience; B.A., Psychology

Leadership & Service

- **President, Career Mobility Committee** (2024–Present) — Graduate organization promoting awareness of diverse career pathways and professional development opportunities.
- **Student Representative, CMB Ph.D. Program Education Committee** (2023–2024) — Contributed to curriculum planning and academic policy discussions.
- Mentored junior trainees in computational analysis, reproducible research practices, and integrative study design bridging experimental and computational approaches.
- Contributed to collaborative manuscript preparation, figure development, and interdisciplinary computational–experimental project design across lab teams.